

ON THE MASS OF FAT ELECTRON

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1. ENERGY CALIBRATION

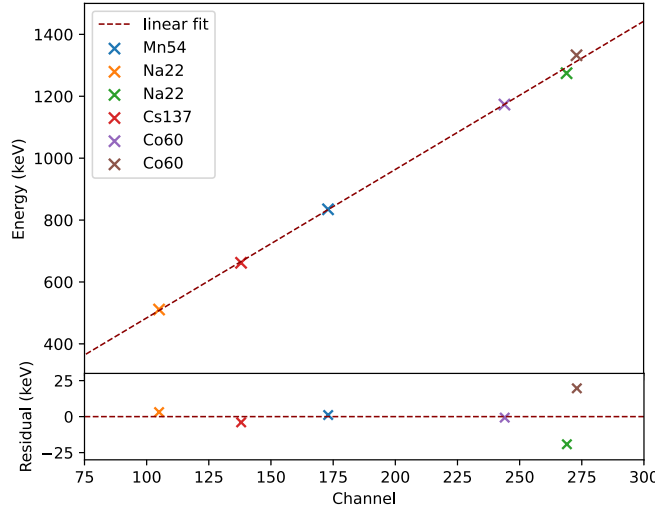


Figure 1. The scattering plot of energy vs. channel, where data points were shown in different colors that correspond to the gamma-ray sources shown in the legend. In addition, the best fit of those data points was also shown in a red dashed line; the residual plot for the fitting is shown at the bottom of the scattering plot. This plot illustrated the linear relationship of the calibration function.

The energy calibration of the multichannel analyzer has been done by observing six peaks from 4 different known gamma-ray spectra. Each of those peaks corresponds to a specific amount of energy shown in Table 1 in appendix. In Table 1, the name of the gamma-ray source is shown in the first column; the channel and the corresponding energy has shown in the second and the last column.

This energy set shows a linear relationship on the top of the Fig. 1, where the energy and channel number are shown on the y-axis and x-axis. The data point for each of those gamma-ray sources has shown in different colors that correspond to the legend on the top left of the figure. In addition, the best-fit line of this data set has shown in a red dotted line, and its residual plot has shown at the bottom of the figure.

The equation of the best fit is given as

$$E = 4.79 \times N + 4.76 \quad (1)$$

where E and N represent the energy and the channel number. This equation was used to calibrate the multichannel analyzer, giving the energy for an arbitrary channel number.

2. DATA

By using six different gamma-ray sources, there are seven sets of the Compton edge data measured from Hackman P216 Lab, where two of those are given by two peaks of ^{22}Na . The data has shown in Table 2.

In Table 2, the gamma-ray sources and their photopeak energies in the unit of keV are given in the first and the second column. The height and depth of counts to the left and right of the edge are given in the third and fourth

columns, and the Compton edge has estimated between those two points. The channel number of the estimated Compton edge and its energy given by the Eq. 1 has shown in the fifth and sixth columns. The uncertainty of the energy given by the calibration function has been estimated to be $\pm 10 \text{ keV}$, which is also shown in the sixth column. The electron's final momentum is given as the equation

$$p c = E_{\gamma} + E_{\gamma'} = E_{\gamma} + (E_{\gamma} - E_{edge}) = 2E_{\gamma} - E_{edge} \quad (2)$$

which pc is given in the last column of Table 2. In addition, the electron's final kinetic energy (K_E) is the same as the Compton edge energy given in the sixth column.

3. RESULT

3.1. Relativistic vs. Newtonian

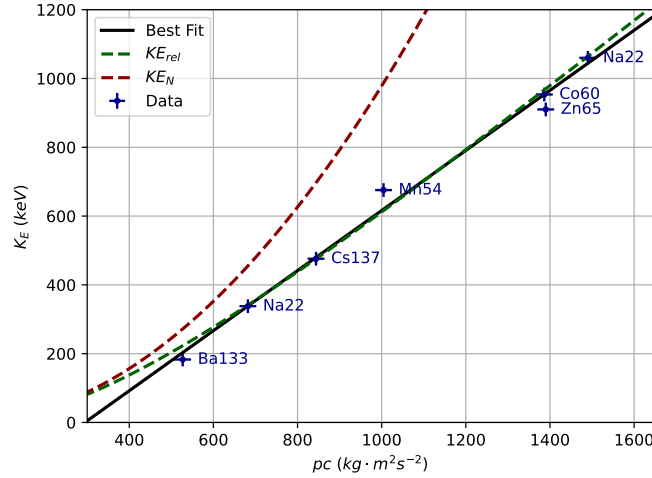


Figure 2. The scattering plot of kinetic energy vs. momentum times the speed of light. The data plots of different gamma-ray sources were shown in blue dots with their uncertainty on both y- and x-direction. The name of each of those sources was labeled right beside each of those data points. The linear best-fit line of those data points was shown in a black solid line. In addition, the predicted values from the relativistic and Newtonian were shown in green and red dashed lines, respectively. This plot shows the relativistic theory has better predicted the data points.

Figure 2 shows a linear relationship between the electron's final kinetic energy (E_{edge}) and the final momentum (pc), where kinetic and momentum times speed of light is given in the y-axis and the x-axis. The data points are shown in blue dots with their uncertainties in y- and x-direction. The corresponding gamma-ray source has been labeled on the right of each data point. The best-fit line of the data set has also given in Fig. 2, which is shown in a black solid line. In addition, the predicted values according to the Newtonian and relativistic theories has shown in red and green dashed lines, respectively.

$$K_{rel} = \sqrt{(mc^2)^2 + (pc)^2} - mc^2 \quad (3)$$

$$K_N = \frac{p^2}{2m} = \frac{(pc)^2}{2mc^2} \quad (4)$$

The equation for the best-fit line is given as

$$K_E = 0.87_{\pm 0.02} \times pc - 257.12_{\pm 23.60} \quad (5)$$

in which the kinetic energy and the momentum times speed of light are given as K_E and pc . This equation has expected to follow either the relativistic (K_{rel}) and Newtonian (K_N) phenomena which are given in Eq. 3 and Eq. 4, respectively. In those equations, the mass of the electron, momentum, and speed of light is given as m , p , and c , respectively. By comparing the predicted value from the relativistic and Newtonian theories shown in Fig. 2, the data is best described by the relativistic theory.

3.2. The value of M_e

According to the relativistic theory, the relationship between the kinetic energy and momentum is given as Eq. 3. The mass of an electron is given as m in the equation. Therefore, the electron mass is obtained by solving m , shown in Eq. 6.

$$M_e = m = \frac{(pc)^2 - K^2}{2c^2 K} \quad (6)$$

Using the Eq. 6, the corresponding values of M_e were calculated and presented in Table 3. The first, second, and third columns give the gamma-ray source, momentum times speed of light, and the final kinetic energy of the electron. The values of the corresponding electron mass with uncertainty are given in the last column. Since these data were found to have huge disagreements, two values of the electron mass $M_e = 865 \pm 152 \text{ keV} \frac{m^2}{s^2}$ and $M_e = 915 \pm 152 \text{ keV} \frac{m^2}{s^2}$ were given from the average and error-weighted average of those values, respectively.

4. DISCUSSION

4.1. Quadratic Calibration Function

Three element examples are selected in the calibration of the matching energy with the corresponding channel. The energies for three elements photopeaks with corresponding channel values are 22Na(511.0 keV), 54Mn(834.8 keV), and 22Na (1274.5 keV). Notice that the 22Na is an element that appeared in 2 photopeaks in the MCA spectrum; one peak is at energy 511.0 keV, and the other is at energy 1274.5 keV. The corresponding channel number of counts are 273 and 664.7, but the photopeak at energy 511keV is selected. To determine the most appropriate fit, the quadratic fit is also considered. However, the quadratic regression line approaches the linear fit. As shown in Figure 3, the quadratic and linear fit were presented in the same plot, and the two are very close. In the quadratic fit equation, the coefficient of N^2 term is $6.74 \times 10^{-4} N^2$, which is approximately zero. Even though the quadratic term is included in the fit analysis, it still is pulled back to the linear relationship.

Therefore, a linear relationship exists between the energy and channel number of counts during the calibration process using the three samples demonstrated above.

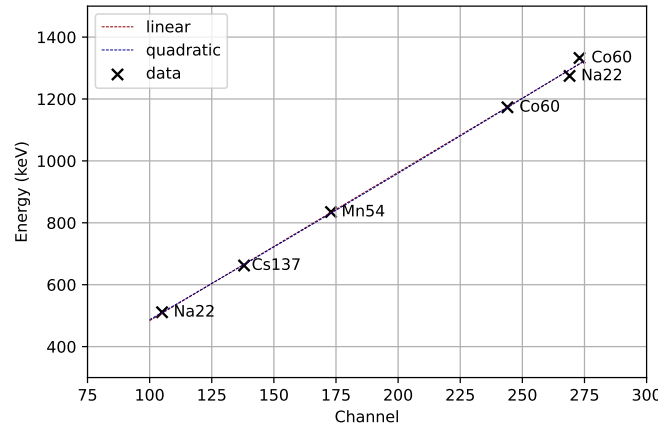


Figure 3. The scattering plot of energy vs. channel, in which data points are plotted in black dots. In addition to a linear best-fit line shown in a red dotted line, a quadratic best-fit line was also shown in a blue dotted line. This plot indicates the result given by the quadratic curve-fitting is barely any different from the one given by linear curve-fitting.

The quadratic fit equation is shown below:

$$E = 6.74 \times 10^{-4} N^2 + 4.53N + 27.15$$

4.2. Comparing with M_e given from previous lab measurements

The value of the electron mass has been precisely measured by [Steiner et al.](#) in the year of 2005. A value of $M_e = 9.11 \times 10^{-31} \text{ kg}$, which is $M_e = 511.0 \text{ keV} \frac{m^2}{s^2}$, was given by their experiments. This value of M_e is significantly smaller than the one given by this lab, even though the uncertainty is considered.

The error-weighted average of the electron mass even more deviates from [Steiner et al.](#)'s value than the one given by the equally weighted average. This has been suspected to be due to biased uncertainties given to the parameters. In table 2, the uncertainty of the Compton edge energy (E_{edge} , the sixth column) has been estimated from an overall uncertainty of calibration function that is $\pm 20 \text{ keV}$. By using the propagation of uncertainty, the electron mass given by a higher magnitude of energy (kinetic energy and momentum) tends to have a smaller uncertainty which, due to the $\pm 20 \text{ keV}$ is a smaller fraction of the measured energy. However, it was found during the calibration that the uncertainty of the energy given by the calibration function tends not to be constant – the uncertainty is getting higher or lower in a specific region of the energy. More specifically, since the gamma-ray sources that have been used to fit the calibration function all fall into the lower energy range, the predicted value in higher energy given by the function has expected to have larger uncertainty than those values in lower energy.

In addition to the problem with the calibration function uncertainty, such a deviation from [Steiner et al.](#)'s values is also caused by the lab measurement. The Compton edge on the spectrum of gamma-ray sources, such as ^{60}Co , is fuzzy, which measures the height and depth of the edge harder. This produced some of the uncertainty that has not been considered in this work and is shown in table 2. This uncertainty would make the interval for the M_e measurement wider and more likely to fall into the range of [Steiner et al.](#)'s values.

Lastly, inspired by Prof. Krebs, since this lab measurement was conducted on Dec. 7, 2022, which is right after the F&M Thanksgiving break, ~~we might get a fat electron who's just coming back from its break.~~ ([Krebs 2022](#))

REFERENCES

Krebs K., 2022, [Mdnaf Namkcah Publication](#), 01, 1

Steiner R. L., Williams E. R., Newell D. B., Liu R., 2005, [Metrologia](#), 42, 431

APPENDIX

Table 1. The calibration data from 4 known gamma ray sources.

Source	Channel	Energy
(–)	(–)	(keV)
⁵⁴ Mn	173	834.8
²² Na	105	511.0
²² Na	269	1274.5
¹³⁷ Cs	138	662.0
⁶⁰ Co	244	1173.2
⁶⁰ Co	273	1332.5

Table 2. The lab measurements of Compton edges from 6 different gamma ray sources.

Source	E_γ	C_1	C_0	N_{edge}	E_{edge}	pc
(–)	(keV)	(counts)	(counts)	(–)	(keV)	($\text{kg} \cdot \text{m}^2 \text{s}^{-2}$)
²² Na	510	35289	9744	22516.5	338.0 ± 20.0	682.0 ± 20.0
²² Na	1275	6446	1503	3974.5	1060.0 ± 20.0	1490.0 ± 20.0
⁵⁴ Mn	840	191	85	138.0	675.57 ± 20.0	1004.43 ± 20.0
¹³³ Ba	355	5028	2362	3695.0	183.0 ± 20.0	527.0 ± 20.0
⁶⁰ Co	1170	1018	545	781.5	953.48 ± 20.0	1386.52 ± 20.0
⁶⁵ Zn	1150	912	141	526.5	910.0 ± 20.0	1390.0 ± 20.0
¹³⁷ Cs	660	765	99	432.0	476.0 ± 20.0	844.0 ± 20.0

Table 3. The electron mass measurements from 6 different gamma ray sources.

Source	pc	K_E	M_e
(–)	($\text{kg} \cdot \text{m}^2 \text{s}^{-2}$)	(keV)	($\text{keV} \cdot \text{m}^2 \text{s}^{-2}$)
Na22	682.0 ± 20.0	338.0 ± 20.0	688 ± 57
Na22	1490.0 ± 20.0	1060.0 ± 20.0	1047 ± 34
Mn54	1004.43 ± 20.0	675.57 ± 20.0	746 ± 37
Ba133	527.0 ± 20.0	183.0 ± 20.0	758 ± 101
Co60	1386.52 ± 20.0	953.48 ± 20.0	1008 ± 35
Zn65	1390.0 ± 20.0	910.0 ± 20.0	1061 ± 38
Cs137	844.0 ± 20.0	476.0 ± 20.0	748 ± 47